

**DEPARTMENT OF
ELECTRONICS & COMMUNICATION ENGINEERING**

PROJECT TITLE: ESP32-Based IoT Anesthesia Monitoring and Assistive Gas Flow Control System

Abstract: The “ESP32-Based IoT Anesthesia Monitoring and Assistive Gas Flow Control System” aims to modernize anesthesia supervision by integrating real-time sensing, data communication, and assistive gas flow regulation into a compact embedded platform. The system measures critical physiological and anesthetic parameters such as SpO₂, EtCO₂, oxygen concentration, pulse rate, and anesthetic gas levels, and uses proportional solenoid valves for controlled gas delivery based on clinician-defined setpoints. IoT connectivity enables continuous data logging, cloud visualization, remote alerts, and tele-support, reducing delays in responding to abnormal physiological trends. By providing threshold-based alarms, decision-support logic, and manual override provisions, the system enhances patient safety while reducing clinician workload. The proposed approach offers a low-cost, modular, and scalable solution suitable for training simulators, research environments, and resource-constrained healthcare facilities.

KEY COMPONENTS: ESP32 Microcontroller, O₂ Anesthetic Gas Sensors, Flow Sensor Pressure Sensor, Pulse Oximeter, Body Temperature Sensor, Display Unit , Alarm & Driver Modules, Power Supply Interface.

Project Batch: BB2

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